

# Narrative Reasoning

Evan Piermont\*      Peio Zuazo-Garin†

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## ABSTRACT

A narrative is accepted only if it is plausible, that is, consistent with an agent’s existing beliefs. We represent an agent’s beliefs through a *stringent implication* relation, capturing the most conservative inferences the agent is willing to draw, and formalize this relation via an evidential structure that assigns each assertion a set of verifying and a set of falsifying states. We show that every relation admits such a representation. Dually, we define a *lenient* implication capturing the weakest inferences consistent with the agent’s beliefs. We show that a narrative is plausible exactly when it lies between these two bounds.

KEYWORDS: Narratives, Implication, Persuasion

This paper investigates how narratives can influence beliefs and subsequently exposes the limits of narrative persuasion. Narratives—rhetorical descriptions of the relationships between distinct situations—allow reasoners to make inferences by organizing events and connections between them. For example, the narrative that “employers cannot afford to pay low-skilled labor any more than they already do” might allow a reasoner to infer that increasing the minimum wage would unambiguously hurt low-skilled workers. Of course, for any narrative to be accepted, it must be plausible in the sense that it is consistent with the reasoner’s existing beliefs. A reasoner who steadfastly believes that businesses drive down wages through monopsony power may reject the narrative above, and as such, may draw different implications from a potential increase in the minimum wage.

To understand the relationship between narratives and beliefs, we put forth a formal model of an agent’s understanding of the logical relationships between different events. From this, we identify the set of narratives that could be deemed plausible by the agent. As such, we characterize the limits of narrative persuasion—the set of

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\*Royal Holloway–University of London, United Kingdom; [evan.piermont@rhul.ac.uk](mailto:evan.piermont@rhul.ac.uk).

†NYU Shanghai, People’s Republic of China; [peio.zuazo.garin@nyu.edu](mailto:peio.zuazo.garin@nyu.edu).

plausible narratives delineate the set of all inferences an agent could be persuaded of without violating his initial beliefs—and make clear the relation between agents’ initial beliefs and how easy it is to persuade him.

The primitive of our model is an agent’s subjective notion of implication representing his most conservative beliefs about the way the world works representing those implications that can be regarded as true on under the highest burden of proof.<sup>1</sup> In particular, we take  $a \Rightarrow b$  to mean that the decision maker regards a lack of evidence falsifying  $a$  to constitute evidence verifying  $b$ .

To understand how this form of implication represents a conservative form of reasoning, consider the following simple example. Let

$a =$  “The restaurant passed its health inspection”

$b =$  “I am willing to eat at the restaurant”

If the condition of the restaurant is the only relevant variable, then  $a$  implies  $b$  in the traditional sense that should  $a$  be true then  $b$  would be true as well. Traditional implication, however, concerns itself with *truth* rather than evidence, and owing to the ‘law of the excluded middle’, assumes that each statement is always either true or false. When regarding evidence, however, a reasoner may encounter a third option, having neither evidence verifying a statement nor evidence falsifying it. For example, the restaurant might put a passing grade in the window (verifying  $a$ ), put a failing grade in the window (falsifying  $a$ ) or have no sign at all.

Our conservative notion of implication (and a reasonable outlook of food safety) then suggests that  $a \not\Rightarrow b$ ; a *lack of evidence* that the restaurant failed is not sufficient to engender patronage.<sup>2</sup>

Interestingly, we show that the set of inferences an agent could be persuaded of is not necessarily transitively closed, so that although an agent might be persuaded that  $a$  implies  $b$  or that  $b$  implies  $c$ , it is not necessarily possible to persuade him that  $a$  implies  $c$ . For example, the United States government seems to have encountered this constraint when, in March 2026, amidst its ongoing war with Iran, it announced it was lifting sanctions on Iranian oil.<sup>3</sup> This strategic move was justified via the

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<sup>1</sup>Piermont and Zuazo-Garin (2026) outline a methodology to elicit subjective implication from choice behavior, which with minor adjustments can be adapted to the present environment.

<sup>2</sup>Contrast this to the negated statements: Here, the conservative reasoning finds “The restaurant failed its health inspection”  $\Rightarrow$  “I am unwilling to eat at the restaurant,” since a lack of a passing grade is itself grounds for dismissal.

<sup>3</sup>No, really, see this New York Times [\[article ↗\]](#).

following two (distinct) narrative arguments. The first is that lifting sanctions on Iranian oil increases global supply, decreasing the price of oil. The second is that high oil prices help Iran, both because it is a net oil exporter and also because high prices create economic pressure on the US.

The various causes and consequences of these two arguments can be arranged as:

$a$  = “The US lift sanctions on Iranian Oil”

$b$  = “The price of oil decreases”

$c$  = “Iran is hurt”

where the first narrative argument attests the implication ‘ $a$  implies  $b$ ’ and the second, ‘ $b$  implies  $c$ .’ One may be tempted then to conclude—as perhaps the US government inclined—that ‘ $a$  implies  $c$ .’ Of course, such an argument is inconsistent, as the implication ‘ $a$  implies  $b$ ’ is predicated on the assumption that Iranian oil exports increase, whereas ‘ $b$  implies  $c$ ’ is predicated on the assumption that Iranian oil exports do not increase.

As such, the US government’s persuasive powers are intransitive; by constructing the proper narrative, the US government can persuade the electorate either that ‘ $a$  implies  $b$ ’ or that ‘ $b$  implies  $c$ ’ but it cannot do both.

## 1 MODEL

Let  $A$  denote a set of assertions (statements about the world) with typical elements  $a, b, c, \dots$ . For any binary relation  $\mathbf{R}$  on  $A$ , i.e., a subset of  $A \times A$ , we write  $a\mathbf{R}b$  to mean  $(a, b) \in \mathbf{R}$ . Say that  $a \in A$  is  $\mathbf{R}$ -null if  $a\mathbf{R}b$  for all  $b \in A$ . For  $\mathbf{R}$  and  $a \in A$ , define  $w_a^{\mathbf{R}} = \{b \in A \mid a\mathbf{R}b\}$ , the ‘ $\mathbf{R}$ -up-set’ of  $a$  and  $m_a^{\mathbf{R}} = \{b \in A \mid b\mathbf{R}a\}$ , the ‘ $\mathbf{R}$ -down-set.’

For any pair of relations  $\mathbf{R}$  and  $\mathbf{S}$  on  $A$ , let  $\mathbf{R}; \mathbf{S}$  denote their composition, that is,  $\mathbf{R}; \mathbf{S} = \{(a, b) \in A \mid \text{there exists } c \in A \text{ such that } (a, c) \in \mathbf{R} \text{ and } (c, b) \in \mathbf{S}\}$ . Note that composition is associative.

## 2 STRINGENT IMPLICATION

The primitive of our model is a binary relation  $\Rightarrow$  defined on  $A$ . We interpret  $\Rightarrow$  as an agent's most conservative beliefs about the way the world works, in the sense that they are the implications that can be regarded as true on under the highest burden of proof. In particular, we think of  $a \Rightarrow b$  to mean that the decision maker regards a lack of evidence falsifying  $a$  to constitute evidence verifying  $b$ . As in the usual sense of implication, should  $a$  be true, the decision maker would believe  $b$  to follow, but here also, the decision maker infers  $b$  even under the weaker condition that  $a$  is simply not known to be false. Stringent implications are therefore indelible beliefs: any further consistent evidence will not overturn them.

To formalize this idea we define a semantic structure embodying this evidential structure:

**DEFINITION 1 (Evidential Structure).** *An **evidential structure** is a triple  $(W, v, f)$  where:  $W$  is a set of states and  $v : A \rightarrow 2^W$  and  $f : A \rightarrow 2^W$  are verification and falsification assignments, respectively. Call an evidential structure*

- **coherent** if  $v(a) \cap f(a) = \emptyset$  for all  $a$ , and,
- **complete** if it is coherent and  $v(a) \cup f(a) = W$  for all  $a$ .

The sates  $v(a)$  represent those states where  $a$  is verified, where there is evidence proving  $a$ , and  $f(a)$  those states where  $a$  is falsified, where there is evidence disproving  $a$ . For an evidential structure  $(W, v, f)$ , let  $v^c(a) = W - v(a)$  and  $f^c(a) = W - f(a)$ , denote those states where  $a$  is *not* verified or falsified, respectively.

These verification and falsification assignments partition the state-space into four regions  $v(a)$  and  $f(a)$ , and then also,  $W - (v(a) \cup f(a))$ , where there is no evidence settling  $a$  in either direction and  $v(a) \cap f(a)$  where there is simultaneously evidence for and against  $a$ . An evidential structure is *coherent* if this fourth possibility never obtains,  $v(a) \cap f(a) = \emptyset$  for all  $a$ , so there is never conflicting evidence in a state. An evidential structure is *complete* if in addition the third possibility never obtains,  $v(a) \cup f(a) = W$  for all  $a$ , so that in every state, every  $a \in A$  is settled as true or false.

We can now state what it means for an evidential structure to represent the stringent implication ' $\Rightarrow$ .' We interpret  $a \Rightarrow b$  to mean that a lack of evidence against

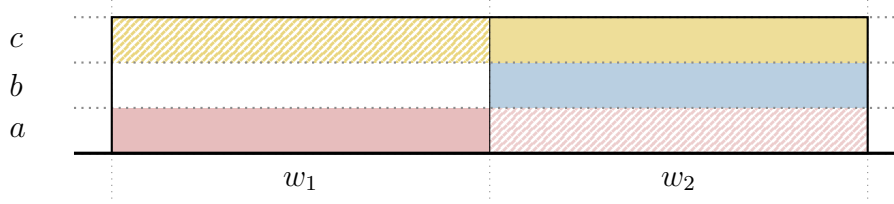


Figure 1:  $v$  is given by solid regions and  $f$  by hashed regions.

$a$  is evidence for  $b$ : if  $w \in f^c(a)$  then  $w \in v(b)$ . Thus, no matter how the truth or falsity of  $a$  and  $b$  get determined in the states where there is currently no evidence,  $a$  will *still* imply  $b$ .

**DEFINITION 2 (Representation).** Say that  $(W, v, f)$  **represents**  $\Rightarrow$  if for all  $a, b \in A$

$$a \Rightarrow b \quad \text{if and only if} \quad f^c(a) \subseteq v(b) \quad (\mathbf{R})$$

Returning to the example in the introduction, we can exhibit a representation for the initial beliefs of the agent:

*Example 1.* Let  $A = a, b, c$  and let  $a \Rightarrow a$ ,  $c \Rightarrow b$ , and  $c \Rightarrow c$ . Take  $W = \{w_1, w_2\}$  and set

$$\begin{aligned} v(a) &= \{w_1\} & f(a) &= \{w_2\} \\ v(b) &= \{w_2\} & f(b) &= \emptyset \\ v(c) &= \{w_2\} & f(c) &= \{w_1\} \end{aligned}$$

Figure 1 shows these assignments, where the solid rectangles represent  $v$  and the hashed,  $f$ . Notice that in  $w_1$ , statement  $b$  is neither verified nor falsified. Within the example, this indicates that in state  $w_1$ , where the US lifts sanctions on Iran, the truth or falsity of this changing oil price is undetermined.

The two narratives set out by the government determine opposite truth assignments: if  $b$  is true at  $w_1$ , lifting sanctions does indeed lower prices ( $a$  implies  $b$ ), but, this is because Iran deemed it advantageous to sell more oil, indicating that  $b$  does not imply  $c$ ; if on the other hand,  $b$  is false at  $w_1$ , lower oil prices are always bad for Iran ( $b$  implies  $c$ ), but, then it would not sell oil just because sanctions are lifted, indicating that  $a$  does not imply  $b$ .

Without further restrictions, state-space  $W$  can be very wasteful, there may be many states that are redundant and could be deleted without alternating the representation. As such, we can demand in addition to the representation property (R) that the  $W$  is minimal in the following sense:

DEFINITION 3 (Tightness). *Call  $(W, v, f)$  a **tight** representation of  $\Rightarrow$  if it is a representation and in addition, for all  $a, b \in A$*

$$(T1) \quad v(a) = \bigcup_{b \in m_a^{\Rightarrow}} f^c(b), \text{ and}$$

$$(T2) \quad f(a) = \bigcup_{b \in w_a^{\Rightarrow}} v^c(b).$$

To understand tightness, notice that by definition of a representation,  $b \in m_a^{\Rightarrow}$  if and only if  $v(a) \supseteq f^c(b)$ , hence the right to left inclusion of (T1) is guaranteed by (R). As such, (T1) is tantamount to requiring that  $v(a)$  is no bigger than it needs to be to satisfy the requirements of the representation—that is it does not include any additional states not included in  $f^c(b)$  for some  $b$  such that  $b \Rightarrow a$ . The requirement (T2) is analogous.

Our notion of representation is flexible enough that *every* implication relation can be represented by a tight evidential structure. Basic logical consistency properties of the implication relation then find counterparts in the structure of the representation:

THEOREM 1. *Every relation  $\Rightarrow$  can be tightly represented by an evidential structure. Moreover,  $\Rightarrow$  can be tightly represented by a coherent (reps., complete) evidential structure if and only if it is transitive (resp., reflexive and transitive).*

🔗 Go to Proof.

### 3 LENIENT IMPLICATION

Dual to stringent implication  $\rightarrow$ , we can define lenient implication as the requires the lowest burden of proof (within the same evidential structure). That is,  $a \rightarrow b$  means that the decision maker regards evidence verifying  $a$  to preclude the falsification of  $b$ . That is, the decision maker may consider it possible that  $a$  is true and  $b$  false, and  $a \rightarrow b$  will still hold as long as  $b$  cannot be *proven to be false* whenever  $a$  is proven to be true.

DEFINITION 4 (Leniency). Call  $\rightarrow$  the **leniency** of  $\Rightarrow$  if:

$$a \rightarrow b \quad \text{if and only if} \quad w_b^{\Rightarrow} \subseteq w_c^{\Rightarrow} \quad \text{for all} \quad c \in m_a^{\Rightarrow} \quad (\text{L})$$

The following are some properties of leniency that are immediate from the definition:

*Remark 1.* If  $\rightarrow$  is the leniency of  $\Rightarrow$ , then:

- (L1)  $\Rightarrow$  is transitive if and only if  $\rightarrow$  is reflexive.
- (L2) If  $\Rightarrow$  is reflexive, then  $\rightarrow$  is transitive.
- (L3) If  $\Rightarrow$  is transitive, then  $\Rightarrow \subseteq \rightarrow$ .
- (L4) If  $\Rightarrow$  is reflexive, then  $\rightarrow \subseteq \Rightarrow$ .
- (L5)  $\rightarrow$  is the largest relation such that  $(\Rightarrow; \rightarrow; \Rightarrow) \subseteq \Rightarrow$ .

Obviously, given  $\Rightarrow$ , the leniency is unique, however, the converse is not true; that is, there are distinct stringent implications that yield the same leniency. Take for example,  $A = \{a, b\}$  and  $\Rightarrow = \emptyset$  and  $\Rightarrow' = (a, b)$ . It is straightforward to check that both of these relations yield the trivial leniency,  $\rightarrow = A \times A$ . Further, the synthesis of (L3) and (L4) shows that if  $\Rightarrow$  is reflexive and transitive, then it coincides with its leniency.

We can now show that the definition of leniency exactly captures the dual relationship between evidence and implication:

THEOREM 2. Fix  $\Rightarrow$ . The following are equivalent:

1.  $\rightarrow$  is the leniency of  $\Rightarrow$ ,
2.  $a \rightarrow b$  iff for every tight representation  $(W, v, f)$ ,  $v(a) \subseteq f^c(b)$ , and,
3.  $a \rightarrow b$  iff for some representation  $(\bar{W}, \bar{v}, \bar{f})$ ,  $\bar{v}(a) \subseteq \bar{f}^c(b)$ .

 **Go to Proof.**

## 4 NARRATIVES

A narrative,  $\mathbf{N}$ , is a reflexive and transitive relation on  $A$ .

DEFINITION 5 (Narrative Plausibility). Fix  $\Rightarrow$  and let  $\rightarrow$  be the (unique) leniency. A narrative is **plausible** for  $\Rightarrow$  if for all  $a, b \in A$ ,

$$a \Rightarrow b \text{ implies } a \mathbf{N} b \quad \text{and} \quad a \mathbf{N} b \text{ implies } a \rightarrow b.$$

Clearly, if  $\Rightarrow \not\subseteq \rightarrow$ , then the set of plausible narratives is empty. Moreover, if  $\Rightarrow = \rightarrow$ , then the only plausible narrative will be  $\Rightarrow$  itself. As such, given Remark 1, the interesting case is principally when  $\rightarrow$  is transitive (so that  $\Rightarrow \subseteq \rightarrow$ ) but not reflexive (so  $\Rightarrow \neq \rightarrow$ ).

THEOREM 3. Then the following are equivalent:

1. A narrative  $\mathbf{N}$  is plausible for  $\Rightarrow$ , and
2. There exists some  $(W, v, f, t)$ , where  $(W, v, f)$  represents  $\Rightarrow$  and  $t : A \rightarrow 2^W$  is such that for all  $a, b \in A$

(P1)  $a \mathbf{N} b$  if and only if  $t(a) \subseteq t(b)$ , and,

(P2)  $v(a) \subseteq t(a) \subseteq f^c(a)$ .

🔗 Go to Proof.

As before, we can define the complement map,  $t^c : a \mapsto W - t(a)$ . Notice then that  $(W, t, t^c)$  defines a *complete* evidential structure for  $\mathbf{N}$ . Since every reflexive and transitive relation can be represented by a complete evidential structure, plausibility's essential requirement is that this complete representation is also consistent with stringent implication  $\Rightarrow$  in the sense that they can be jointly represented by a single evidential structure.

## 5 AGGREGATING NARRATIVES

Consider a set  $I = \{1, \dots, n\}$  of *experts*; each  $i \in I$  entertains a narrative  $\mathbf{N}_i$ . When the expert's are not in complete agreement, a reasoner may want to aggregate the information held by each expert individually. Although the experts' narratives are all complete reflexive and transitive, and so by Theorem 1, can be understood as *complete* evidential structures, disagreement between them will preclude maintaining

this precision in the aggregate.<sup>4</sup>

**THEOREM 4.** *Let  $\{n_i\}_{i \in I}$  be a set of expert narratives and  $\Rightarrow$  be a stringent implication. The following are equivalent:*

1.  $\Rightarrow \subseteq \bigcap_{i \in I} \mathbf{N}_i$  and for all  $i$ ,  $(\Rightarrow; \mathbf{N}_i; \Rightarrow) \subseteq \Rightarrow$
2. For all  $i \in I$ ,  $\mathbf{N}_i$  is plausible for  $\Rightarrow$ .

**THEOREM 5.** *Let  $\{n_i\}_{i \in I}$  be a set of expert narratives and  $\Rightarrow$  be a stringent implication. The following are equivalent:*

1. For all  $a, b, c \in A$ , and  $i, j \in I$ ,  $a \Rightarrow b$  if and only if  $c\mathbf{N}_i a$  implies  $c\mathbf{N}_j b$
2. There exists some  $(W, v, f, \{t_i\}_{i \in I})$ , where  $(W, v, f)$  represents  $\Rightarrow$ ,  $(W, t_i, t_i^c)$  represents  $\mathbf{N}_i$  and for all  $a \in A$ ,  $v(a) = \bigcap_{i \in I} t_i(a)$  and  $f(a) = \bigcap_{i \in I} t_i^c(a)$ .

 **Go to Proof.**

**COROLLARY 6.** *Let  $\Rightarrow$  be the stringent implication derived from  $\{n_i\}_{i \in I}$ . Then for all  $i \in I$ ,  $\mathbf{N}_i$  is plausible for  $\Rightarrow$ .*

## A PROOFS

All proofs have been formalized and verified by the Lean programming language, see [\[project files ↗\]](#) and [\[interactive web version ↗\]](#).

*Proof of Theorem 1.* Take the state-space to be  $A$  itself and define

$$f(a) = \{z \in A \mid w_a^{\Rightarrow} \not\subseteq w_z^{\Rightarrow}\} \tag{F}$$

$$v(a) = m_a^{\Rightarrow} \tag{V}$$

We will prove that  $(W, v, f)$  forms a tight representation:

**(R)** Let  $a \Rightarrow b$ . We must show  $f^c(a) \subseteq v(b)$ . So let  $z \in f^c(a)$ ; by **(F)**,  $w_a^{\Rightarrow} \subseteq w_z^{\Rightarrow}$ .

Since  $a \Rightarrow b$ , we have  $b \in w_a^{\Rightarrow} \subseteq w_z^{\Rightarrow}$ , or,  $z \in w_b^{\Rightarrow}$ . Thus, by **(V)**,  $z \in v(b)$ .

Conversely, let  $a \not\Rightarrow b$ . We must show  $f^c(a) \not\subseteq v(b)$ . By **(F)**,  $a \in f^c(a)$ . Since

$a \not\Rightarrow b$ ,  $a \notin m_b^{\Rightarrow}$ , and therefore by **(V)**,  $a \notin v(b)$ .

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<sup>4</sup>Note that a naive form of aggregation would be to simply take the intersection of each narrative as  $\mathbf{N} = \bigcap_{i \in I} \mathbf{N}_i \subseteq A \times A$ . The result is a reflexive and transitive ordering.

- (T1) Fix  $a \in A$ . We will show that  $v(a) \subseteq \bigcup_{b \in m_a^{\rightarrow}} f^c(b)$  as the opposite inclusion is guaranteed by (R). Let  $z \in v(a)$ . By (V),  $z \in m_a^{\rightarrow}$ . Moreover, since  $w_z^{\rightarrow} \subseteq w_z^{\rightarrow}$ , (F) implies  $z \in f^c(z) \subseteq \bigcup_{b \in m_a^{\rightarrow}} f^c(b)$ .
- (T2) Fix  $a \in A$ . By taking the complement of both sides, it suffices to establish:  $f^c(a) = \bigcap_{b \in w_a^{\rightarrow}} v(b)$ . We will show that  $\bigcap_{b \in w_a^{\rightarrow}} v(b) \subseteq f^c(a)$ , as the opposite inclusion is guaranteed by (R). Let  $z \in \bigcap_{b \in w_a^{\rightarrow}} v(b)$ . Let  $x \in w_a^{\rightarrow}$ , we have  $z \in v(x)$ , or by (V),  $x \in w_z^{\rightarrow}$ . Since  $x$  was arbitrary, it follows that  $w_a^{\rightarrow} \subseteq w_z^{\rightarrow}$ , or by (F),  $z \in f^c(a)$ .

We can now show the further properties:

1. **Coherency.** Let  $\Rightarrow$  be transitive; so,  $w_a^{\rightarrow} \subseteq w_b^{\rightarrow}$  for all  $b \in m_a^{\rightarrow}$ . Hence, by (T2) we have  $f^c(b) \subseteq f^c(a)$  for all such  $b$ . Thus by (T1), we have  $v(a) = \bigcup_{b \in m_a^{\rightarrow}} f^c(b) \subseteq f^c(a)$ , as needed. The other direction is obvious.
2. **Completeness.** In addition, let  $\Rightarrow$  be reflexive and let  $b \in f^c(a)$ . So,  $w_a^{\rightarrow} \subseteq w_b^{\rightarrow}$ . By reflexivity  $a \in w_a^{\rightarrow}$ , and hence  $a \in w_b^{\rightarrow}$  and  $b \in v(a)$ . ■

*Proof of Theorem 2. (1 implies 2)* Let  $(W, v, f)$  be any tight representation for  $\Rightarrow$ .

Let  $a \rightarrow b$ , we must show that  $v(a) \subseteq f^c(b)$ . If  $v(a) = \emptyset$ , this holds vacuously, so assume that  $w \in v(a)$ . By (T1),  $w \in \bigcup_{z \in m_a^{\rightarrow}} f^c(z)$ ; so there exist some  $x \in m_a^{\rightarrow}$ , with  $w \in f^c(x)$ . Since  $a \rightarrow b$ , (L) insists that  $w_b^{\rightarrow} \subseteq w_x^{\rightarrow}$ . Thus by (T2),  $f(b) \subseteq f(x)$ :  $w \in f^c(x) \subseteq f^c(b)$ , as desired.

Now, assume that  $a \not\rightarrow b$ , we must show that  $v(a) \not\subseteq f^c(b)$ . By the contrapositive of (L), there exists some  $c \in m_a^{\rightarrow}$  such that  $w_b^{\rightarrow} \not\subseteq w_c^{\rightarrow}$ . Let  $d \in w_b^{\rightarrow} - w_c^{\rightarrow}$ . Thus we have by (R),  $f^c(b) \subseteq v(d)$  and  $f^c(c) \not\subseteq v(d)$ . So let  $w \in f^c(c) - v(d)$ ; by (T1),  $w \in v(a)$ .

(2 implies 3) The only if direction follows immediately from the existence of a tight representation as given by Theorem 1. To see the if direction, fix some  $(W, v, f)$  that tightly represents  $\Rightarrow$  and assume that  $(\bar{W}, \bar{v}, \bar{f})$  that represents  $\Rightarrow$  and such that  $\bar{v}(a) \subseteq \bar{f}^c(b)$ . We then have that:

$$\bigcup_{c \in m_a^{\rightarrow}} \bar{f}^c(c) \subseteq \bar{v}(a) \subseteq \bar{f}^c(b) \subseteq \bigcap_{d \in w_b^{\rightarrow}} \bar{v}(d).$$

By (R) it follows that  $c \Rightarrow d$  for every  $c \in m_a^{\rightarrow}$  and  $d \in w_b^{\rightarrow}$ . As  $(W, v, f)$  represents

$\Rightarrow$ , we then have that:

$$v(a) = \bigcup_{c \in m_a^{\vec{\Rightarrow}}} f^c(c) \subseteq \bigcap_{d \in w_b^{\vec{\Rightarrow}}} v(d) = f^c(b),$$

which from our assumption of (2), we can conclude that  $a \rightarrow b$ .

(3 implies 1) Assume  $a \rightarrow b$ . So there exists some  $(\bar{W}, \bar{v}, \bar{f})$  that represents  $\Rightarrow$  and  $\bar{v}(a) \subseteq \bar{f}^c(b)$ . Let  $c \in m_a^{\vec{\Rightarrow}}$ , so that  $\bar{f}^c(c) \subseteq \bar{v}(a) \subseteq \bar{f}^c(b)$ . By (R),  $w_b^{\vec{\Rightarrow}} \subseteq w_c^{\vec{\Rightarrow}}$ . The forward direction of (L) holds.

Next, assume that for all  $c \in m_a^{\vec{\Rightarrow}}$  we have  $w_b^{\vec{\Rightarrow}} \subseteq w_c^{\vec{\Rightarrow}}$ . Let  $(W, v, f)$  be a tight representation of  $\Rightarrow$ , which exists by Theorem 1. By (T2) implies  $f(c) \subseteq f(b)$ . So by (T1),  $v(a) = \bigcup_{c \in m_a^{\vec{\Rightarrow}}} f(c) \subseteq f(b)$ . From our assumption of (3), we can conclude that  $a \rightarrow b$ . The reverse direction of (L) holds. ■

*Proof of Remark 1.*

(L1) Assume  $\Rightarrow$  is transitive. Let  $c \in m_a^{\vec{\Rightarrow}}$  and  $x \in w_a^{\vec{\Rightarrow}}$ ; so  $c \Rightarrow a \Rightarrow x$ . By the transitivity of  $\Rightarrow$ ,  $x \in w_c^{\vec{\Rightarrow}}$ . This establishes  $w_a^{\vec{\Rightarrow}} \subseteq w_c^{\vec{\Rightarrow}}$ , and hence  $a \rightarrow a$  by (L). So  $\rightarrow$  is reflexive.

Now, assume  $\rightarrow$  is reflexive. Let  $c \Rightarrow a$  and  $a \Rightarrow b$ ; so  $c \in m_a^{\vec{\Rightarrow}}$  and  $b \in w_a^{\vec{\Rightarrow}}$ . By reflexivity,  $a \rightarrow a$ , and so by (L),  $b \in w_a^{\vec{\Rightarrow}} \subseteq w_c^{\vec{\Rightarrow}}$ , meaning  $c \Rightarrow b$ . Thus,  $\Rightarrow$  is transitive.

(L2) Assume  $\Rightarrow$  is reflexive. Let  $a \rightarrow b$  and  $b \rightarrow d$ . Since  $b \rightarrow d$  and  $b \Rightarrow b$ , meaning  $b \in m_b^{\vec{\Rightarrow}}$ , (L) yields  $w_d^{\vec{\Rightarrow}} \subseteq w_b^{\vec{\Rightarrow}}$ . Moreover, since  $a \rightarrow b$ , for any  $c \in m_a^{\vec{\Rightarrow}}$ , (L) yields  $w_d^{\vec{\Rightarrow}} \subseteq w_b^{\vec{\Rightarrow}} \subseteq w_c^{\vec{\Rightarrow}}$ . Thus,  $a \rightarrow d$ .

(L3) Assume  $\Rightarrow$  is transitive, and let  $a \Rightarrow b$ . Let  $c \in m_a^{\vec{\Rightarrow}}$ , so  $c \Rightarrow x$ . Let  $x \in w_b^{\vec{\Rightarrow}}$ , so  $b \Rightarrow x$ . By transitivity  $c \Rightarrow x$ , establishing  $w_b^{\vec{\Rightarrow}} \subseteq w_c^{\vec{\Rightarrow}}$ . Therefore,  $a \rightarrow b$ .

(L4) Assume  $\Rightarrow$  is reflexive, and let  $a \rightarrow b$ . Because  $\Rightarrow$  is reflexive,  $a \in m_a^{\vec{\Rightarrow}}$ , so by (L),  $w_b^{\vec{\Rightarrow}} \subseteq w_a^{\vec{\Rightarrow}}$ . Again by reflexivity,  $b \in w_b^{\vec{\Rightarrow}} \subseteq w_a^{\vec{\Rightarrow}}$ . Therefore  $a \Rightarrow b$ .

(L5) First, we show that  $\rightarrow$  satisfies  $(\Rightarrow; \rightarrow; \Rightarrow) \subseteq \Rightarrow$ . Let  $(c, d) \in (\Rightarrow; \rightarrow; \Rightarrow)$ . This means there exist  $a, b \in A$  such that  $c \Rightarrow a$ ,  $a \rightarrow b$ , and  $b \Rightarrow d$ . Because  $a \rightarrow b$ , (L) implies  $w_b^{\vec{\Rightarrow}} \subseteq w_c^{\vec{\Rightarrow}}$ . We know  $b \Rightarrow d$ , so  $d \in w_b^{\vec{\Rightarrow}} \subseteq w_c^{\vec{\Rightarrow}}$ . Thus  $c \Rightarrow d$ .

Finally, we show  $\rightarrow$  is the largest such relation. Suppose **R** is any relation satisfying  $(\Rightarrow; \mathbf{R}; \Rightarrow) \subseteq \Rightarrow$ . We will show  $\mathbf{R} \subseteq \rightarrow$ . Let  $a \mathbf{R} b$ . Let  $c \in m_a^{\vec{\Rightarrow}}$  and  $d \in w_b^{\vec{\Rightarrow}}$ . Then

we have  $(c, d) \in (\Rightarrow; \mathbf{R}; \Rightarrow) \subseteq \Rightarrow$ ; it follows that  $d \in w_c^{\Rightarrow}$ . Since  $d$  was arbitrary,  $w_b^{\Rightarrow} \subseteq w_c^{\Rightarrow}$ , hence by (L),  $a \rightarrow b$ .  $\blacksquare$

*Proof of Theorem 3. (2 implies 1)* Let  $(W, f, v, t)$  be the joint representation as given by (2). First assume  $a \Rightarrow b$ . By (P2) and (R), we have  $t(a) \subseteq f^c(a) \subseteq v(b) \subseteq t(b)$ ; by (P1),  $a \mathbf{N} b$ . Now, assume  $a \mathbf{N} b$ , so (P1) and (P2) we have  $v(a) \subseteq t(a) \subseteq t(b) \subseteq f^c(b)$ . Hence, by Theorem 2,  $a \rightarrow b$ .

*(1 implies 2)* Let  $(\bar{W}, \bar{v}, \bar{f})$  be a tight representation of  $\Rightarrow$ , which exists by Theorem 1. Define  $(W, f, v, t)$  as follows:

- $W = \bar{W} \sqcup A$  (i.e., is the disjoint union of  $\bar{W}$  and some (disjoint) copy of  $A$ )
- $v : a \mapsto \bar{v}(a) \sqcup m_a^{\mathbf{N}}$
- $f : a \mapsto \bar{f}(a) \sqcup (A - m_a^{\mathbf{N}})$
- $t : a \mapsto \bigcup_{c \in m_a^{\mathbf{N}}} \bar{v}(c) \sqcup m_a^{\mathbf{N}}$

We will now show that  $(W, f, v, t)$  satisfies the conditions required by (2):

- (R) Let  $a \Rightarrow b$  and  $w \in f^c(a) = (\bar{f}^c(a) \sqcup m_a^{\mathbf{N}})$ . Then either (i)  $w \in \bar{f}^c(a)$ , in which case by (R), we have  $w \in \bar{v}(b) \subseteq v(b)$ , or (ii)  $w \in m_a^{\mathbf{N}}$ , so  $w \mathbf{N} a$ . Since  $\mathbf{N}$  is plausible,  $a \Rightarrow b$  implies  $a \mathbf{N} b$ , and the transitivity of  $\mathbf{N}$  then delivers  $w \mathbf{N} b$ , or  $w \in m_b^{\mathbf{N}} \subseteq v(b)$ . Hence  $f^c(a) \subseteq v(b)$ . Now assume that  $a \not\Rightarrow b$ . By (R) there exists some  $w \in \bar{W}$  such that  $w \in \bar{f}^c(a) - \bar{v}(b)$ , and hence  $w \in f^c(a) - v(b)$ ;  $f^c(a) \not\subseteq v(b)$ .
- (P1) Let  $a \mathbf{N} b$ . So by transitivity  $m_a^{\mathbf{N}} \subseteq m_b^{\mathbf{N}}$ ; it is immediate then, that  $t(a) \subseteq t(b)$ . Now let  $t(a) \subseteq t(b)$ . By reflexivity,  $a \in m_a^{\mathbf{N}} \subseteq t(a)$ , so it follows that  $a \in m_b^{\mathbf{N}}$ . By definition,  $a \mathbf{N} b$ .
- (P2) By reflexivity,  $a \in m_a^{\mathbf{N}}$ , so  $\bar{v}(a) \subseteq \bigcup_{c \in m_a^{\mathbf{N}}} \bar{v}(c)$ ; it follows that  $v(a) \subseteq t(a)$ . Now, let  $w \in t(a)$ ; it is immediate that if  $w \in m_a^{\mathbf{N}}$  then  $w \in f^c(a)$ , so assume that  $w \in \bar{v}(c)$  for some  $c \in m_a^{\mathbf{N}}$ . Since  $\mathbf{N}$  is plausible,  $c \mathbf{N} a$  implies  $c \rightarrow a$ . By Theorem 2, and the tightness of  $(\bar{W}, \bar{v}, \bar{f})$ , we have  $\bar{v}(c) \subseteq \bar{f}^c(a)$ . Again, we have  $w \in \bar{f}^c(a) \subseteq f^c(a)$ . Hence,  $t(a) \subseteq f^c(a)$ .  $\blacksquare$

*Proof of Theorem 5. (1 implies 2).* Define the state space  $(W, \{t_i\}_{i \in I})$  as  $W = A$  and for each  $i \in I$  let  $t_i : a \mapsto m_a^{\mathbf{N}_i}$ . Clearly,  $(W, t_i, t_i^c)$  represents  $\mathbf{N}_i$ . So define

$$v(a) := a \mapsto \bigcap_{i \in I} t_i(a) \quad \text{and} \quad f(a) := a \mapsto \bigcap_{i \in I} t_i^c(a),$$

and we will show that  $(W, v, f)$  represents  $\Rightarrow$ .

Let  $a \Rightarrow b$ . If  $f^c(a) = \emptyset$  then the inclusion is obvious, so assume  $z \in f^c(a) = (A - \bigcap_{i \in I} t_i^c(a)) = \bigcup_{i \in I} t_i(a)$ . So we have that  $z \in t_i(a)$ , i.e.,  $z \mathbf{N}_i a$ , for some  $i \in I$ . By (1), this implies that for all  $j \in I$ ,  $z \mathbf{N}_j b$ , i.e.,  $z \in t_j(b)$ . Thus,  $z \in \bigcap_{j \in I} t_j(b) = v(b)$ .

Now assume  $f^c(a) \subseteq v(b)$ . Let  $i \in I$  and  $z \in A$  be such that  $z \mathbf{N}_i a$  (if no such pair exists, then the RHS of (1) holds vacuously, and so  $a \Rightarrow b$ ). Then  $z \in t_i(a) \subseteq \bigcup_{k \in I} t_k(a) = f^c(a) \subseteq v(b) = \bigcap_{j \in I} t_j(b)$ . So  $z \mathbf{N}_j b$  for all  $j \in I$ . Thus by (1), we have  $a \Rightarrow b$ .

(2 implies 1) Let  $a \Rightarrow b$  and assume for some  $i \in I$  and  $z \in A$ ,  $z \mathbf{N}_i a$ . Thus appealing to the multi-representation given by (2), we have  $t_i(z) \subseteq t_i(a) \bigcup_{k \in I} t_k(a) = f^c(a) \subseteq v(b) = \bigcap_{j \in I} t_j(b)$ . Thus for all  $j \in I$ ,  $z \mathbf{N}_j b$ .

Now fix  $a, b$  and assume that for all  $i, j \in I$  and  $z \in A$ ,  $z \mathbf{N}_i a$  implies  $z \mathbf{N}_j b$ . If  $f^c(a) = \emptyset$  then  $a \Rightarrow b$  immediately via (R), so assume  $z \in f^c(a) = \bigcup_{i \in I} t_i(a)$ . So we have that  $z \in t_i(a)$ , i.e.,  $z \mathbf{N}_i a$ , for some  $i \in I$ . By our assumption, this implies that for all  $j \in I$ ,  $z \mathbf{N}_j b$ , i.e.,  $z \in t_j(b)$ . Thus,  $z \in \bigcap_{j \in I} t_j(b) = v(b)$ . So  $f^c(a) \subseteq v(b)$ , and by (R),  $a \Rightarrow b$ . ■

## REFERENCES

**Piermont, Evan and Peio Zuazo-Garin**, “Failures of Contingent Thinking,” 2026.